JW6 Labs 2 and 3

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Instructions and time-saving hints. The purpose of this lab is to learn linear regression in R. You will reproduce the commands blocks in JW Section 6.5 in this Rmarkdown file. **Please do not retype all the R commands as shown in Section 6.5.** Instead, go to link to text, find Chapter 6 Lab text, and cut and paste the text into your Rmarkdown file. Then break up the R commands into the R chunks shown in the text. **Do not put all commands into the same block!**

After breaking the text into the correct chunks, you may knit the document. You will notice some problems when knitting. This is due to the fact that the pasted commands are meant for the command line, not Rmarkdown. You must find Rmarkdown conversions for these problems. One problem is the appearance of the `fix()` commands in the blocks, which displays dataframes in a different window. As a substitute for `fix()`, please use the `kable` command from library `knitr`, as in `knitr::kable()`. The results, however, will be too long to hand in! Use this only for your personal use. For your homework/lab submission, please instead use `head()` instead of `fix()`. The TA will be less burdened that way.

Since R chunks are independent pieces of code, you will sometimes produce errors when imitating the format in the book. For example, the `abline()` commands must be grouped in the same chunk as the initial `plot()` command.

Congratulations! When the document is successfully knitted, you should read Section 6.5 along with the Rmarkdown output. You should see the commands along with the output plots. Together, JW 6.5 and your Rmarkdown file will help you to learn linear regression in R. Note that the plotting is done with core R commands and not with the `ggplot2` package. Submit your Rmd file along with an unzipped PDF of the result before the deadline.

Chapter 6 Lab 2: Ridge Regression and the Lasso

```
x=model.matrix(Salary~.,Hitters)[,-1]
y=Hitters$Salary
```

Ridge Regression

```
library(glmnet)
```

```
## Warning: package 'glmnet' was built under R version 4.3.2
```

```
## Loading required package: Matrix
```

```
## Loaded glmnet 4.1-8
```

```r
grid=10^seq(10,-2,length=100)
ridge.mod=glmnet(x,y,alpha=0,lambda=grid)
```

```r
dim(coef(ridge.mod))
```

```
## [1]  20 100
```

```r
ridge.mod$lambda[50]
```

```
## [1] 11497.57
```

```r
coef(ridge.mod)[,50]
```

```
##   (Intercept)         AtBat          Hits         HmRun          Runs 
## 407.356050200   0.036957182   0.138180344   0.524629976   0.230701523 
##           RBI         Walks         Years        CAtBat         CHits 
##   0.239841459   0.289618741   1.107702929   0.003131815   0.011653637 
##        CHmRun         CRuns          CRBI        CWalks       LeagueN 
##   0.087545670   0.023379882   0.024138320   0.025015421   0.085028114 
##     DivisionW       PutOuts       Assists        Errors    NewLeagueN 
##  -6.215440973   0.016482577   0.002612988  -0.020502690   0.301433531
```

```r
sqrt(sum(coef(ridge.mod)[-1,50]^2))
```

```
## [1] 6.360612
```

```r
ridge.mod$lambda[60]
```

```
## [1] 705.4802
```

```r
coef(ridge.mod)[,60]
```

```
##  (Intercept)        AtBat         Hits        HmRun         Runs          RBI 
##  54.32519950   0.11211115   0.65622409   1.17980910   0.93769713   0.84718546 
##        Walks        Years       CAtBat        CHits       CHmRun        CRuns 
##   1.31987948   2.59640425   0.01083413   0.04674557   0.33777318   0.09355528 
##         CRBI       CWalks      LeagueN    DivisionW      PutOuts      Assists 
##   0.09780402   0.07189612  13.68370191 -54.65877750   0.11852289   0.01606037 
##       Errors   NewLeagueN 
##  -0.70358655   8.61121213
```

```r
sqrt(sum(coef(ridge.mod)[-1,60]^2))
```

```
## [1] 57.11001
```

```r
predict(ridge.mod,s=50,type="coefficients")[1:20,]
```

```
##   (Intercept)         AtBat          Hits         HmRun          Runs
##  4.876610e+01 -3.580999e-01  1.969359e+00 -1.278248e+00  1.145892e+00
##           RBI         Walks         Years        CAtBat         CHits
##  8.038292e-01  2.716186e+00 -6.218319e+00  5.447837e-03  1.064895e-01
##        CHmRun         CRuns          CRBI        CWalks       LeagueN
##  6.244860e-01  2.214985e-01  2.186914e-01 -1.500245e-01  4.592589e+01
##     DivisionW       PutOuts       Assists        Errors    NewLeagueN
## -1.182011e+02  2.502322e-01  1.215665e-01 -3.278600e+00 -9.496680e+00
```

```r
set.seed(1)
train=sample(1:nrow(x), nrow(x)/2)
test=(-train)
y.test=y[test]
```

```r
ridge.mod=glmnet(x[train,],y[train],alpha=0,lambda=grid, thresh=1e-12)
ridge.pred=predict(ridge.mod,s=4,newx=x[test,])
mean((ridge.pred-y.test)^2)
```

```
## [1] 142199.2
```

```r
mean((mean(y[train])-y.test)^2)
```

```
## [1] 224669.9
```

```r
ridge.pred=predict(ridge.mod,s=1e10,newx=x[test,])
mean((ridge.pred-y.test)^2)
```

```
## [1] 224669.8
```

```r
ridge.pred=predict(ridge.mod,s=0,newx=x[test,],exact=T,x=x[train,],y=y[train])
mean((ridge.pred-y.test)^2)
```

```
## [1] 168588.6
```

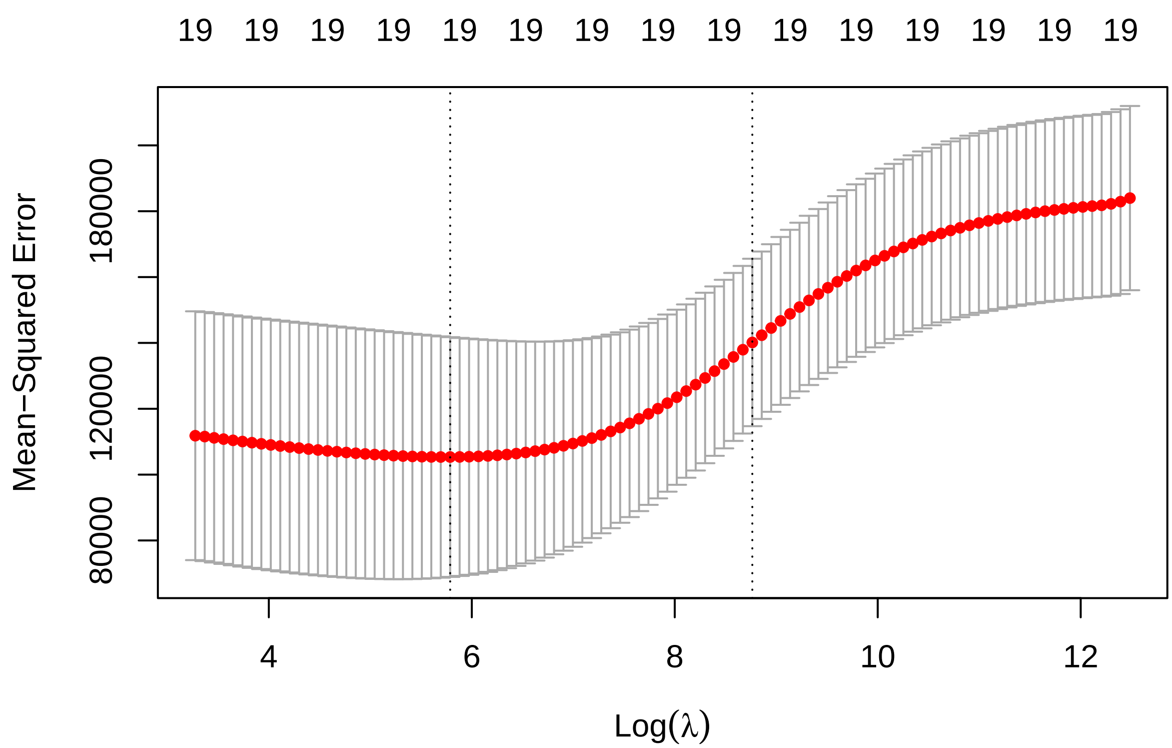
```r
lm(y~x, subset=train)
```

```
## 
## Call:
## lm(formula = y ~ x, subset = train)
## 
## Coefficients:
## (Intercept)       xAtBat        xHits       xHmRun        xRuns         xRBI
##    274.0145      -0.3521      -1.6377       5.8145       1.5424       1.1243
##      xWalks       xYears      xCAtBat       xCHits      xCHmRun       xCRuns
##      3.7287     -16.3773      -0.6412       3.1632       3.4008      -0.9739
##       xCRBI      xCWalks     xLeagueN   xDivisionW     xPutOuts     xAssists
##     -0.6005       0.3379     119.1486    -144.0831       0.1976       0.6804
##     xErrors  xNewLeagueN
##     -4.7128     -71.0951
```

```r
predict(ridge.mod,s=0,exact=T,type="coefficients",x=x[train,],y=y[train])[1:20,]
```

```
##  (Intercept)        AtBat         Hits        HmRun         Runs          RBI
##  274.0200994   -0.3521900   -1.6371383    5.8146692    1.5423361    1.1241837
##        Walks        Years       CAtBat        CHits       CHmRun        CRuns
##    3.7288406  -16.3795195   -0.6411235    3.1629444    3.4005281   -0.9739405
##         CRBI       CWalks      LeagueN    DivisionW      PutOuts      Assists
##   -0.6003976    0.3378422  119.1434637 -144.0853061    0.1976300    0.6804200
##       Errors   NewLeagueN
##   -4.7127879  -71.0898914
```

```r
set.seed(1)
cv.out=cv.glmnet(x[train,],y[train],alpha=0)
plot(cv.out)
```




```r
bestlam=cv.out$lambda.min
bestlam
```

```
## [1] 326.0828
```

```r
ridge.pred=predict(ridge.mod,s=bestlam,newx=x[test,])
mean((ridge.pred-y.test)^2)
```

```
## [1] 139856.6
```

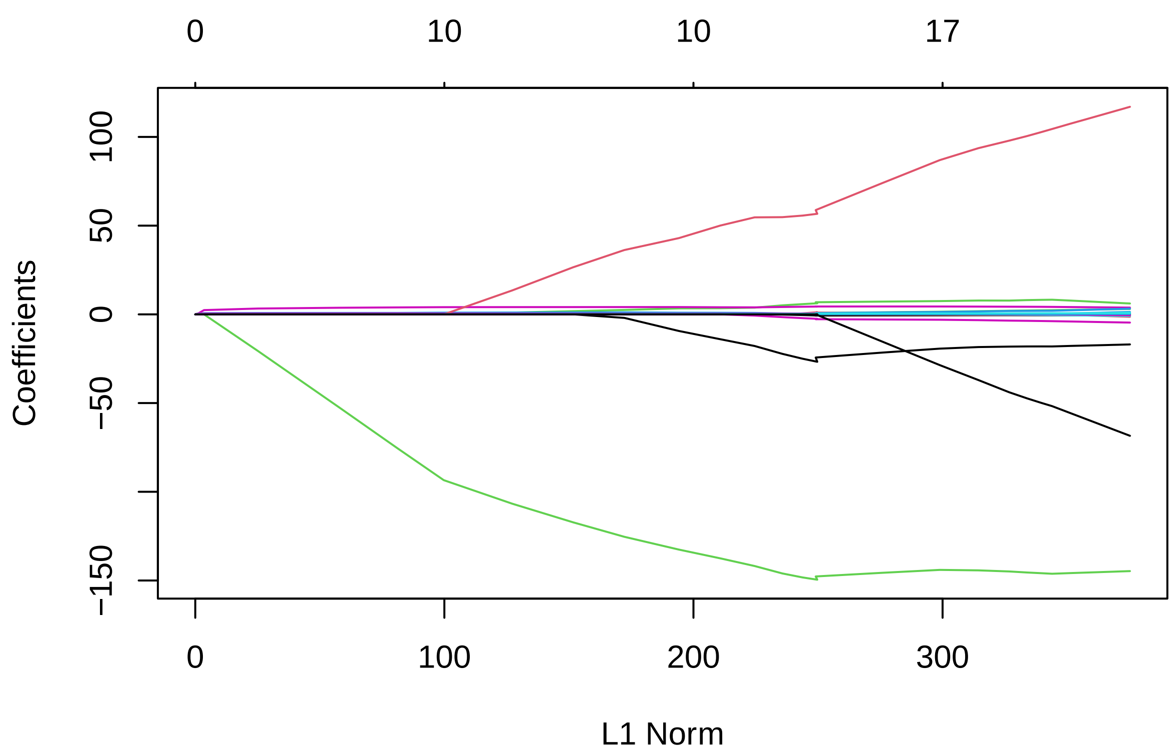
```r
out=glmnet(x,y,alpha=0)
predict(out,type="coefficients",s=bestlam)[1:20,]
```

```
##  (Intercept)        AtBat         Hits        HmRun         Runs          RBI
##  15.44383120   0.07715547   0.85911582   0.60103106   1.06369007   0.87936105
##        Walks        Years       CAtBat        CHits       CHmRun        CRuns
##   1.62444617   1.35254778   0.01134999   0.05746654   0.40680157   0.11456224
##         CRBI       CWalks      LeagueN    DivisionW      PutOuts      Assists
##   0.12116504   0.05299202  22.09143197 -79.04032656   0.16619903   0.02941950
##       Errors   NewLeagueN
##  -1.36092945   9.12487765
```

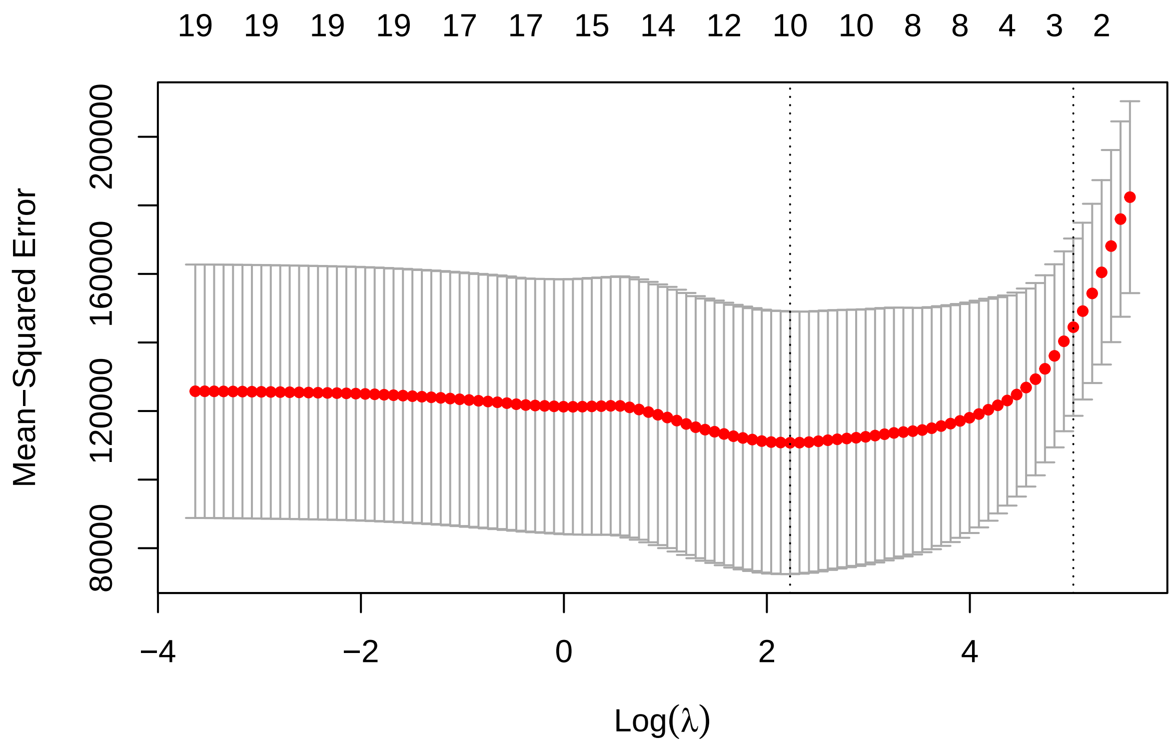
The Lasso

```r
lasso.mod=glmnet(x[train,],y[train],alpha=1,lambda=grid)
plot(lasso.mod)
```

```
## Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm):
## collapsing to unique 'x' values
```



```r
set.seed(1)
cv.out=cv.glmnet(x[train,],y[train],alpha=1)
plot(cv.out)
```




```r
bestlam=cv.out$lambda.min
lasso.pred=predict(lasso.mod,s=bestlam,newx=x[test,])
mean((lasso.pred-y.test)^2)
```

```
## [1] 143673.6
```

```r
out=glmnet(x,y,alpha=1,lambda=grid)
lasso.coef=predict(out,type="coefficients",s=bestlam)[1:20,]
lasso.coef
```

```
##   (Intercept)         AtBat          Hits         HmRun          Runs
##    1.27479059   -0.05497143    2.18034583    0.00000000    0.00000000
##           RBI         Walks         Years        CAtBat         CHits
##    0.00000000    2.29192406   -0.33806109    0.00000000    0.00000000
##        CHmRun         CRuns          CRBI        CWalks       LeagueN
##    0.02825013    0.21628385    0.41712537    0.00000000   20.28615023
##     DivisionW       PutOuts       Assists        Errors    NewLeagueN
## -116.16755870    0.23752385    0.00000000   -0.85629148    0.00000000
```

```r
lasso.coef[lasso.coef!=0]
```

```
##   (Intercept)         AtBat          Hits         Walks         Years 
##    1.27479059   -0.05497143    2.18034583    2.29192406   -0.33806109 
##        CHmRun         CRuns          CRBI       LeagueN     DivisionW 
##    0.02825013    0.21628385    0.41712537   20.28615023 -116.16755870 
##       PutOuts        Errors 
##    0.23752385   -0.85629148
```

Chapter 6 Lab 3: PCR and PLS Regression

Principal Components Regression

```r
library(pls)
```

```
## Warning: package 'pls' was built under R version 4.3.3
```

```
## 
## Attaching package: 'pls'
```

```
## The following object is masked from 'package:stats':
## 
##     loadings
```

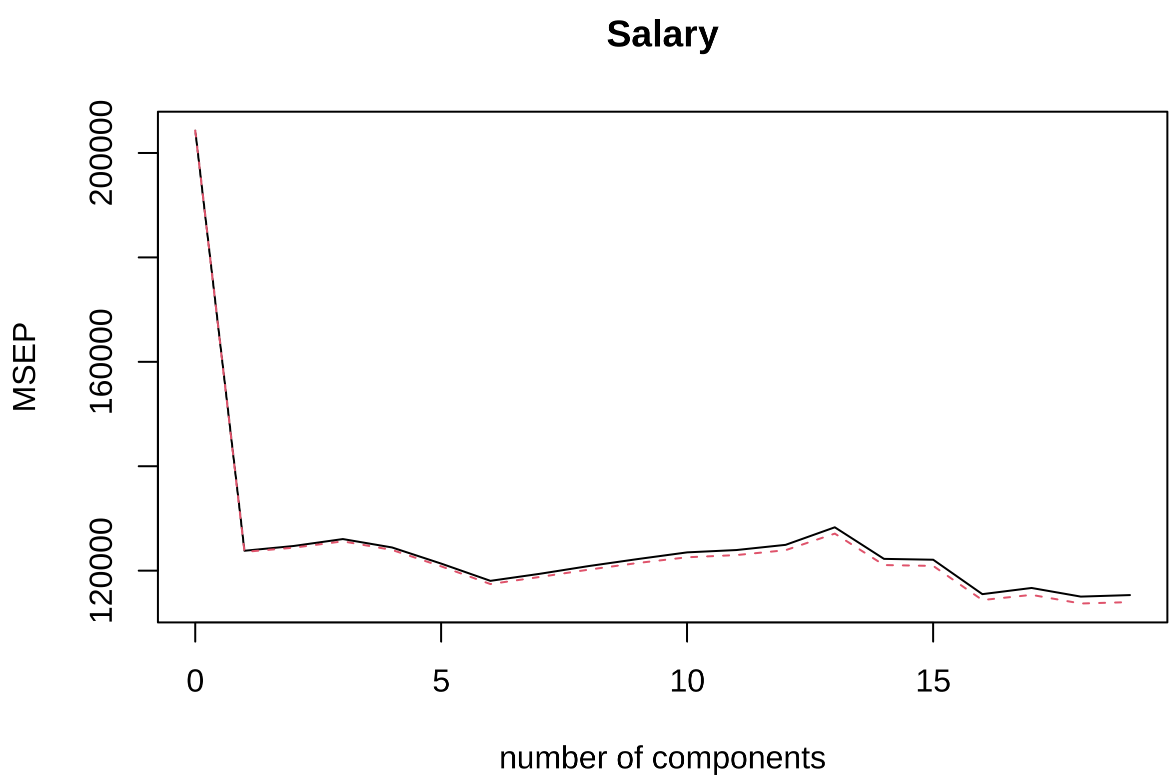
```r
set.seed(2)
pcr.fit=pcr(Salary~., data=Hitters,scale=TRUE,validation="CV")
```

```r
summary(pcr.fit)
```

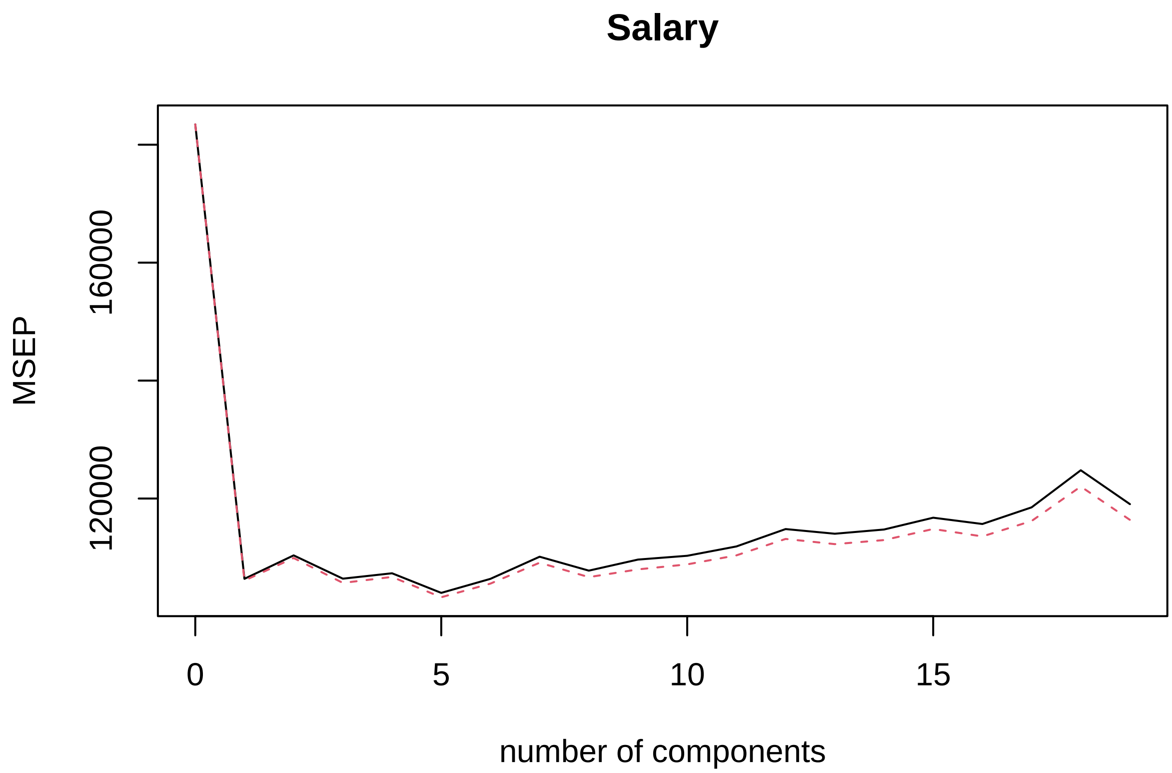
```
## Data:    X dimension: 263 19 
##  Y dimension: 263 1
## Fit method: svdpc
## Number of components considered: 19
## 
## VALIDATION: RMSEP
## Cross-validated using 10 random segments.
##        (Intercept)  1 comps  2 comps  3 comps  4 comps  5 comps  6 comps
## CV             452    351.9    353.2    355.0    352.8    348.4    343.6
## adjCV          452    351.6    352.7    354.4    352.1    347.6    342.7
##        7 comps  8 comps  9 comps  10 comps  11 comps  12 comps  13 comps
## CV       345.5    347.7    349.6     351.4     352.1     353.5     358.2
## adjCV    344.7    346.7    348.5     350.1     350.7     352.0     356.5
##        14 comps  15 comps  16 comps  17 comps  18 comps  19 comps
## CV        349.7     349.4     339.9     341.6     339.2     339.6
## adjCV     348.0     347.7     338.2     339.7     337.2     337.6
## 
## TRAINING: % variance explained
```

```
##          1 comps  2 comps  3 comps  4 comps  5 comps  6 comps  7 comps  8 comps
## X          38.31    60.16    70.84    79.03    84.29    88.63    92.26    94.96
## Salary     40.63    41.58    42.17    43.22    44.90    46.48    46.69    46.75
##          9 comps  10 comps  11 comps  12 comps  13 comps  14 comps  15 comps
## X          96.28     97.26     97.98     98.65     99.15     99.47     99.75
## Salary     46.86     47.76     47.82     47.85     48.10     50.40     50.55
##          16 comps  17 comps  18 comps  19 comps
## X           99.89     99.97     99.99    100.00
## Salary      53.01     53.85     54.61     54.61
```

```r
validationplot(pcr.fit,val.type="MSEP")
```




```r
set.seed(1)
pcr.fit=pcr(Salary~., data=Hitters,subset=train,scale=TRUE, validation="CV")
validationplot(pcr.fit,val.type="MSEP")
```




```
pcr.pred=predict(pcr.fit,x[test,],ncomp=7)
mean((pcr.pred-y.test)^2)
```

```
## [1] 140751.3
```

```
pcr.fit=pcr(y~x,scale=TRUE,ncomp=7)
summary(pcr.fit)
```

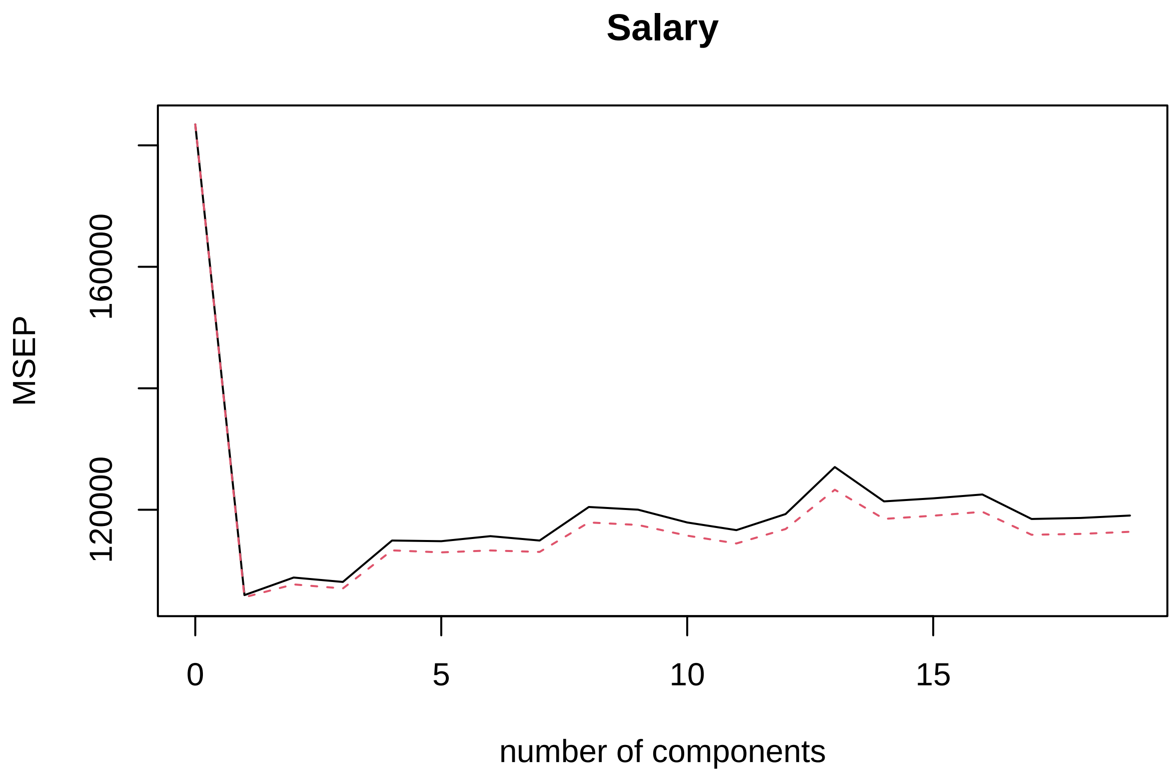
```
## Data:    X dimension: 263 19
##  Y dimension: 263 1
## Fit method: svdpc
## Number of components considered: 7
## TRAINING: % variance explained
##    1 comps  2 comps  3 comps  4 comps  5 comps  6 comps  7 comps
## X    38.31    60.16    70.84    79.03    84.29    88.63    92.26
## y    40.63    41.58    42.17    43.22    44.90    46.48    46.69
```

Partial Least Squares

```
set.seed(1)
pls.fit=plsr(Salary~., data=Hitters,subset=train,scale=TRUE, validation="CV")
summary(pls.fit)
```

```
## Data:    X dimension: 131 19
##  Y dimension: 131 1
## Fit method: kernelpls
## Number of components considered: 19
##
## VALIDATION: RMSEP
## Cross-validated using 10 random segments.
##        (Intercept)  1 comps  2 comps  3 comps  4 comps  5 comps  6 comps
## CV           428.3    325.5    329.9    328.8    339.0    338.9    340.1
## adjCV        428.3    325.0    328.2    327.2    336.6    336.1    336.6
##        7 comps  8 comps  9 comps  10 comps  11 comps  12 comps  13 comps
## CV       339.0    347.1    346.4     343.4     341.5     345.4     356.4
## adjCV    336.2    343.4    342.8     340.2     338.3     341.8     351.1
##        14 comps  15 comps  16 comps  17 comps  18 comps  19 comps
## CV        348.4     349.1     350.0     344.2     344.5     345.0
## adjCV     344.2     345.0     345.9     340.4     340.6     341.1
##
## TRAINING: % variance explained
##         1 comps  2 comps  3 comps  4 comps  5 comps  6 comps  7 comps  8 comps
## X         39.13    48.80    60.09    75.07    78.58    81.12    88.21    90.71
## Salary    46.36    50.72    52.23    53.03    54.07    54.77    55.05    55.66
##         9 comps  10 comps  11 comps  12 comps  13 comps  14 comps  15 comps
## X         93.17     96.05     97.08     97.61     97.97     98.70     99.12
## Salary    55.95     56.12     56.47     56.68     57.37     57.76     58.08
##         16 comps  17 comps  18 comps  19 comps
## X          99.61     99.70     99.95    100.00
## Salary     58.17     58.49     58.56     58.62
```

```r
validationplot(pls.fit,val.type="MSEP")
```




```
pls.pred=predict(pls.fit,x[test,],ncomp=2)
mean((pls.pred-y.test)^2)
```

```
## [1] 145367.7
```

```
pls.fit=plsr(Salary~., data=Hitters,scale=TRUE,ncomp=2)
summary(pls.fit)
```

```
## Data:    X dimension: 263 19
##  Y dimension: 263 1
## Fit method: kernelpls
## Number of components considered: 2
## TRAINING: % variance explained
##         1 comps  2 comps
## X         38.08    51.03
## Salary    43.05    46.40
```